

Opinion

Neurofeedback through the lens of reinforcement learning

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Despite decades of experimental and clinical practice, the neuropsychological mechanisms underlying neurofeedback (NF) training remain obscure. NF is a unique form of reinforcement learning (RL) task, during which participants are provided with rewarding feedback regarding desired changes in neural patterns. However, key RL considerations – including choices during practice, prediction errors, credit-assignment problems, or the exploration–exploitation tradeoff – have infrequently been considered in the context of NF. We offer an RL-based framework for NF, describing different internal states, actions, and rewards in common NF protocols, thus fashioning new proposals for characterizing, predicting, and hastening the course of learning. In this way we hope to advance current understanding of neural regulation via NF, and ultimately to promote its effectiveness, personalization, and clinical utility.

NF as a special kind of RL task

NF is a collection of brain–computer interface procedures through which individuals learn to regulate their own neural dynamics by receiving reinforcing feedback regarding desired changes in neural activity/connectivity patterns (Figure 1A) [1,2]. Guided by the feedback, regulation learning is measured through the change in neural patterns in the reinforced direction (Figure 1B). NF protocols differ with respect to various task characteristics [3]. However, the two most critical dimensions that define structurally different learning regimes are the goals of regulation, as instructed to participants before practice, and the timing of feedback presentation (Figure 1C).

The goals of regulation may be specified through explicit instructions or through an implicit protocol (Figure 1C, y axis). In an explicit regulation protocol, participants are asked to actively search in a trial-and-error manner for mental strategies (eliciting some kind of mental content involving perceptual, imaginative, cognitive, or metacognitive components) that result in a defined neural change (e.g., the upregulation or downregulation of a neural activity) [4,5]. Sensory feedback reporting the change then informs participants about their corresponding reward. During ‘pure’ implicit NF practice, reward is similarly provided when the neural change is in line with a desired neural pattern. However, participants are neither informed about its neural origin nor provided with explicit instructions for regulation [6]. Intermediate protocols on the spectrum from explicit to implicit NF have also been examined [7,8], where partial information is provided to participants regarding the goals of neural regulation. Accordingly, in the ensuing discussion on the pros and cons of explicitness of regulation, such cases may resemble more explicit rather than implicit protocols, depending on the specific instructions given.

The second dimension is the timing of feedback presentation (Figure 1C, x axis). Sensory feedback may be given either during or following periods of neural regulation attempts. Most NF

Highlights

In neurofeedback (NF) participants learn to regulate their own neural dynamics via rewarding feedback regarding desired changes in neural patterns.

As NF is a form of reinforcement learning (RL) task, it can be beneficial to systematically analyze it in these terms.

We offer an RL-based theoretical account for NF, describing the various states, actions, and rewards in common NF protocols.

In this opinion article we delineate known RL issues (such as credit assignment problems and exploration–exploitation tradeoff) as manifested in NF.

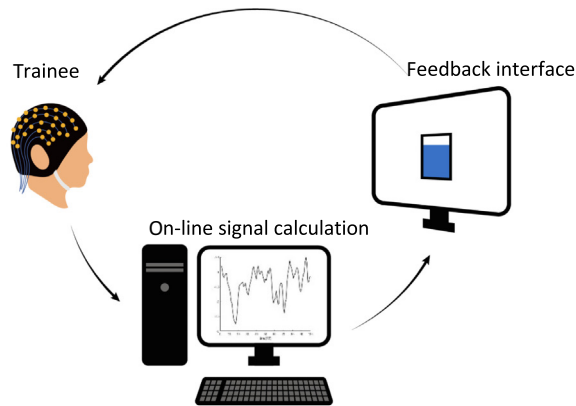
Representational learning and metacognition play several incompletely understood roles in NF, and action selection merits more attention. We propose specific methodological changes that reflect the involvement and significance of these processes.

We propose RL algorithms for characterizing and predicting NF learning. It is hoped that the proposed framework will promote NF as a basic science tool, and potentially as an efficient intervention for brain and mental disorders.

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(A) Closed-loop neurofeedback training set-up

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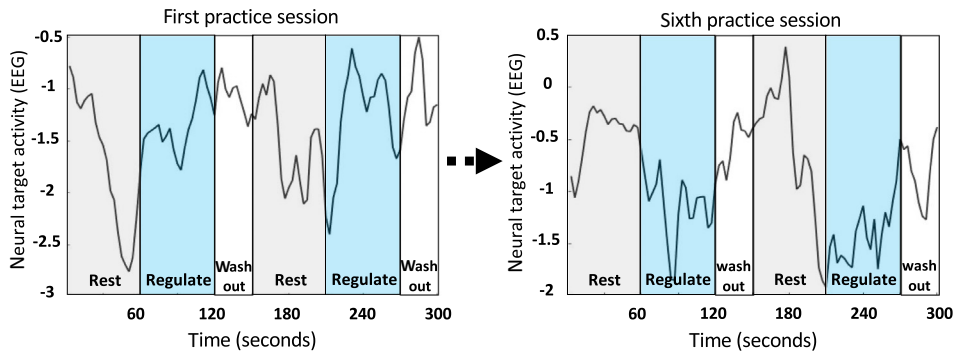
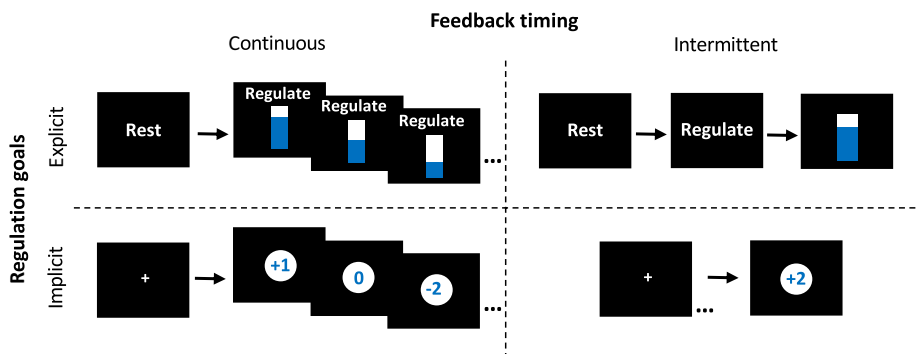
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(B) Neurofeedback learning: development of regulation abilities**(C) Neurofeedback protocol dimensions**

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Figure 1. Neurofeedback (NF) training. (A) Closed-loop neurofeedback training set-up. An individual's neural activity is probed via a recording device – for example, functional magnetic resonance imaging (fMRI), electroencephalogram (EEG), functional near-infrared spectroscopy (fNIRS) – calculated in real time and presented back to the trainee through a sensory (e.g., visual, auditory) feedback interface, thus closing the loop. Based on task instructions and contingent sensory feedback, the trainee learns to self-regulate their own neural patterns. (B) NF learning: development of regulation abilities. Neural activity of an amygdala-related electrical fingerprint of a representative trainee practicing an explicit-continuous protocol (for details and group-level results see [5]). During 'rest' phases, participants were instructed to attend passively to a screen. During 'regulate' phases, participants were instructed to downregulate their neural target activity using some mental strategy, in face of continuous feedback. During 'washout' phases, participants performed a simple filler task of finger tapping. In this

(Figure legend continued at the bottom of the next page.)

protocols provide continuous sensory feedback regarding changes in neural patterns during regulation attempts (e.g., [9]). The ultimate reward is a function of this sensory feedback (for example, if the reported neural change is appropriate for at least a minimal period of time). However, in intermittent protocols, feedback presentation is either delayed or is completely separated from the time of regulation [10,11].

Various settings along these two dimensions (explicit/implicit, and continuous/intermittent) may have unique advantages and constraints, such as dealing with noisy data or directing subjects towards specific regulation strategies (for a more detailed discussion, see [12]). Here, we focus on particular implications of these two dimensions of NF. For convenience, we distinguish three broad regimes: ‘explicit-continuous’, where regulation and feedback presentation are simultaneous (i.e., an online report of the utility of regulation attempts), ‘explicit-intermittent’, where regulation and feedback presentation are temporally dissociated, and an ‘implicit’ (continuous or intermittent) NF protocol. Indeed, different NF protocols may lead to different learning dynamics [10,15,78].

In line with the description given earlier, it is generally acknowledged that NF is a form of a RL task in which agents learn which (in this case, internal) actions to take based on reinforcement given by the environment [13]. Previous studies have provided important supporting evidence for the involvement of RL parameters in specific contexts of NF training (e.g., [14–23]). However, key known aspects of RL – such as participants’ behavioral choices and exploration during learning, the predictions they make regarding their choices’ outcomes (reward expectancies), and the resulting prediction errors – are not commonly documented during practice or considered in analyses of NF. Consequently, the involvement of such underlying parameters in NF learning have yet to be systematically established.

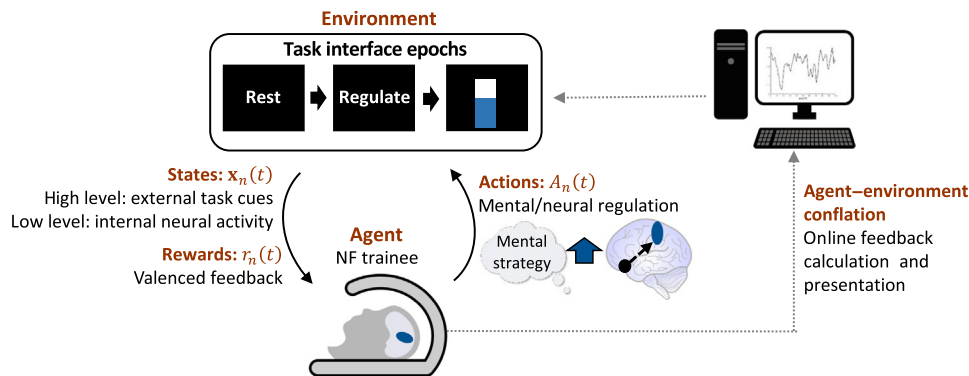
This opinion article has two primary goals. The first is to suggest a general RL-based account for NF learning. This proposed framework can help us to understand and evaluate the underlying parameters that might direct neural regulation learning via NF. The second goal is to use this account to fashion new algorithmic suggestions for the different NF protocol types which could be used both for characterizing learning and, potentially, for improving the efficiency of NF protocols.

RL modeling of NF: articulating the framework

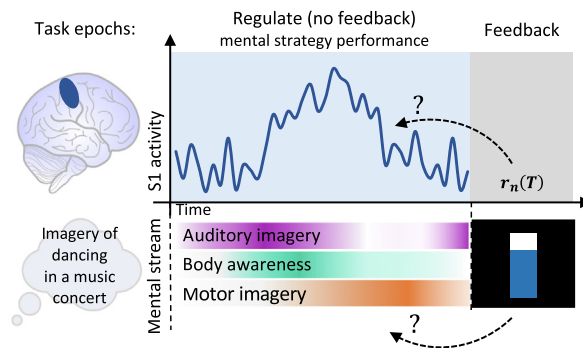
RL generically models environments which involve three elements: states, actions, and rewards (Figure 2A). In this opinion article, the state at time t on trial n will be noted as $x_n(t)$; in classic RL tasks, the state could report, for example, on the position of an animal in a maze, or, in NF practice, on an external condition such as the command to ‘regulate’, or on the internal state of the brain. Actions will be denoted by $A_n(t)$; for example, movements in the maze, or, in NF, the choice to imagine something in particular, or the neural process that gives rise to the reinforced neural pattern in a targeted brain region. Lastly, rewards, which depend on states and actions, will be denoted, for example, by $r_n(t)$; the reward could be a consummatory

case, lower activity during a regulation phase relative to the previous rest phase indicates a change in the reinforced direction, and vice versa. Accordingly, the left panel demonstrates two unsuccessful NF cycles from the first practice session, and the right panel shows two successful NF cycles from the last (sixth) practice session. (C) NF protocol dimensions. Different settings along two main task characteristics that dictate structurally different learning regimes: instructed regulation goals, which may be explicit (e.g., ‘try to make the neural feedback increase using some mental strategy’) or implicit (e.g., ‘attend to the rewarding stimuli appearing on the screen’), and feedback timing, which may be provided continuously (presented for each measured time point) during regulation attempts, or intermittently (presented following regulation periods).

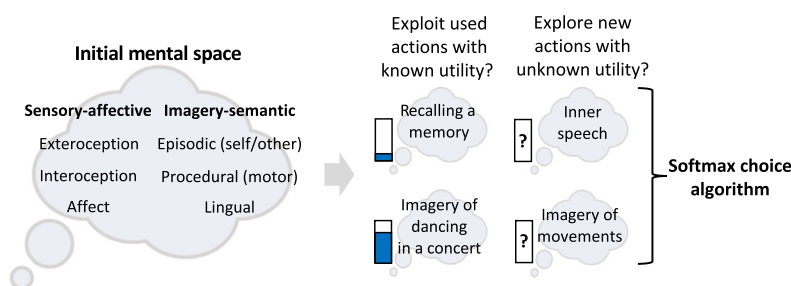
(A) Reinforcement learning elements in neurofeedback



(B) Credit assignment problems in neurofeedback



(C) Strategy choice and exploration-exploitation tradeoff in neurofeedback



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Figure 2. Reinforcement learning (RL) framework for neurofeedback (NF) training. (A) RL elements in NF. In NF, the environment consists of the task interface, which reports to the agent (the NF trainee) on high-level states (generally noted as $x_n(t)$ at time t on trial n), which are the changing task cues (e.g., ‘rest’, ‘regulate’), and of observations that are evaluated as rewards $r_n(t)$. The agent performs mental/neural actions $A_n(t)$ that influence the transitions between lower-level neural states – the internal neural activity of different regions in the brain [also noted as $x_n(t)$], which in turn changes the feedback. This closed-loop nature of NF paradigms conflates the agent with the environment. The implications of this conflation issue are discussed in the main text. (B) Credit assignment problems in neurofeedback. During regulation epochs in an explicit-intermittent protocol, the trainee applies a mental strategy (e.g., imagery of dancing in a music concert) which has several simultaneous mental features of differing amplitudes. Following regulation, feedback of aggregated activity level is presented [$r_n(T)$], which faces the trainee with figuring out which mental features were specifically related to regulation success. (C) Strategy choice and exploration–exploitation tradeoff in neurofeedback. Trainees have to balance between choosing actions (mental strategies) with known utility that were employed before, and exploring new options that have yet been practiced. This process can be described algorithmically by the softmax choice rule (Box 3).

object (e.g., a sweet liquid), or in NF a positive feedback in response to required changes in neural patterns.

Actions change the state of the environment, for example, according to dynamics that can be formulated as follows:

$$\mathbf{x}_n(t+1) = f[\mathbf{x}_n(t), \mathbf{A}_n(t)] + \varepsilon_n^x(t) \quad [1]$$

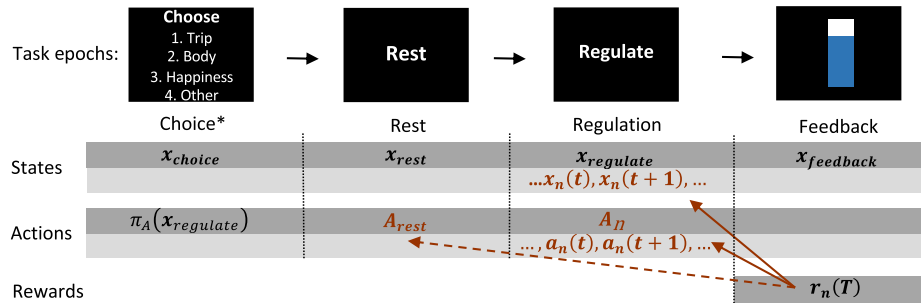
where f specifies the deterministic component of the transitions, and $\varepsilon_n^x(t)$ is state noise. For convenience, we generally consider continuous states and discrete time steps [e.g., the repetition time (TR) in functional magnetic resonance imaging (fMRI)]; however, the conclusions would also apply to continuous time.

Agents learn a *policy* [$\pi(\mathbf{x}_n(t))$], which is a systematic mapping of states to actions, or perhaps a probability distribution over actions. The policy indicates which states and actions should maximize long-term rewards, for example, rewards accumulated across the whole trial. Agents are faced with both structural credit assignment problems (working out which aspect of the state is relevant for and/or indicative of reward) and temporal credit assignment problems (awarding responsibility for a delayed reward or lack thereof to an earlier action that was responsible) (Figure 2B). Critical to many facets of RL is a value function [$V^\pi(\mathbf{x}_n(t))$], which estimates the sum total reward (called the return) available from a particular state following the current policy π , or an action-value function [$Q^\pi(\mathbf{x}_n(t), \mathbf{A}_n(t))$] which estimates the return from that state given a particular first action and then following the policy. Learning value functions is useful, since states with high values are good targets even without immediate reward, since they promise good reward in the future. In the context of NF, this will be particularly important when feedback is delayed (as in explicit-intermittent protocols).

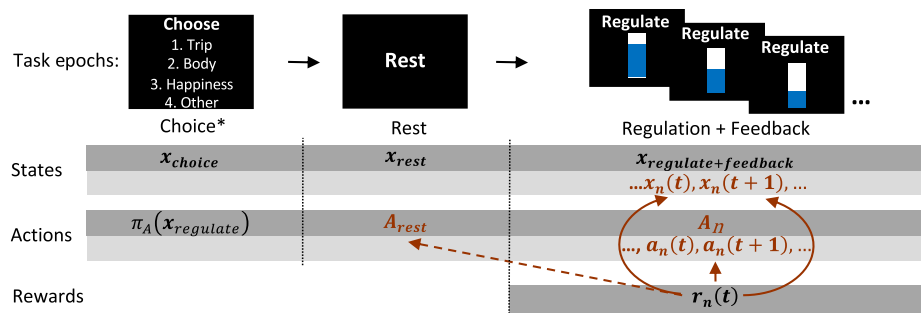
A special issue for NF in terms of RL is the conflation of agent and environment. In RL, these two elements are traditionally separate, and communicate only through actions, observations (which report on states), and rewards. There is a recent trend towards treating rewards as subjective, and seeing them as generated internally based on environmental inputs, rather than being objective and generated directly by the environment [24,25]. However, this is exacerbated in NF, as the same brain/mind that is responsible for choosing and performing an action is also the brain/mind whose internal states are intended to be changed by that action, or by the mere presentation of the contingent feedback (Figure 2A). For example, consider asking participants to downregulate activity that normally reports predictions of future reinforcement (e.g., neural activity in the ventral striatum). Successful downregulation would be associated with a positive prediction, which would subvert that downregulation. This amounts to a form of what is called Pavlovian misbehavior in other contexts [26]. While the possible involvement of Pavlovian (classical) conditioning in NF is discussed below (see section ‘RL algorithms for NF training’), we will generally ignore this problem and imagine that the experimenter is providing feedback regarding activity in brain areas that are not associated with solving the underlying RL problem itself.

We turn next to consider states, actions, and rewards in the coarse NF protocol categories articulated above: explicit-intermittent, explicit-continuous, and implicit (see Figure 3 for a demonstration of the RL accounts per NF protocol type). Based on this articulation, we hope to shed light on how known RL elements – such as value functions, policies, credit assignment problems

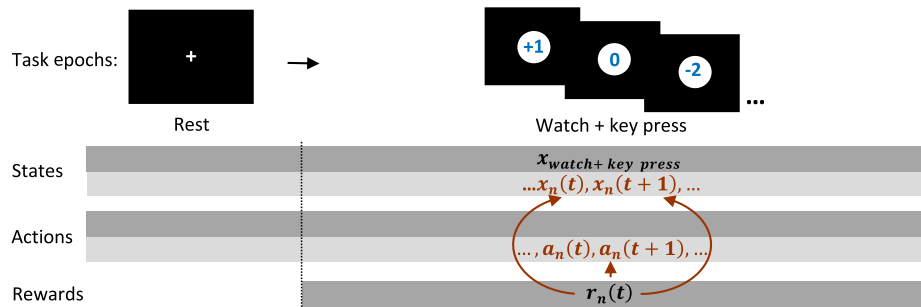
(A) Explicit - intermittent



(B) Explicit - continuous



(C) Implicit



Conscious mental stream Unconscious neural/mental stream $A_n / \uparrow$ Evaluative processes

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Figure 3. States, actions, and rewards in neurofeedback (NF) training protocols. Each protocol type – (A) explicit-intermittent, (B) explicit-continuous, and (C) implicit – is considered with respect to the learning environment that is arranged by its consecutive task epochs. The choice screen is a new proposition we put forward that disentangles the choice state from other task states (detailed in the main text), during which participants can choose which mental strategy to employ. The task epochs determine the various reinforcement learning (RL) elements: the states the agent will pass through, denoted for example as $x_{regulate}$ for high-level representations of task epochs, and as $x_n(t)$ for low level unconscious mental/neural states; the conditions under which the agent will perform actions, denoted by A_n for conscious mental strategies and by $a_n(t)$ for the unconscious causal precursors of the reinforced neural pattern, that function as actions; and the rewards the agent will receive, denoted in various protocol-specific ways [e.g., $r_n(t)$]. These different RL elements are manifested consciously (dark gray) and unconsciously (pale gray). Evaluative processes are colored in brown.

(Figure 2B), and exploration-exploitation tradeoffs (Figure 2C) – are manifested in different NF protocols.

States: from the representation of task epochs to fine-grained mental/neural events

As a result of the conflation between agent and environment, in NF the state $x_n(t)$ has at least two hierarchical components. At the higher level is the component that is arranged by the environment, that is, by the consecutive protocol conditions that are observed by the participant, such as ‘rest’ and ‘regulate’ (denoted by x_{rest} and x_{regulate} , respectively). Constrained by these external task cues, at the lower level is the state component of internal, detailed activity in different areas of the brain (Figure 3A–C, States). In NF, the dynamics described in Equation 1 will apply to transitions between the low-level internal states rather than the high-level representations of task epochs.

Learning about the prospective value of these states (and actions) takes place both consciously and unconsciously (as illustrated in Figure 3). Specifically, some aspects of the lower level of the state – the unconscious stream of mental and neural activity – might be subject to metacognitive awareness, and therefore become consciously represented and afford conscious control through mental strategy application (Box 1). This could be useful if, for example, it is beneficial for regulation for the brain to be in a macroscopic state, such as calmness, that is internally detectable. However, even in explicit protocols, other parts of the low-level internal states of the brain might just be available to participate in an implicit, unconscious, learning process which can determine state values and policies (Box 3). This is aligned with RL paradigms, which involve both putatively more conscious processes (e.g., model-based decision-making) [27] and unconscious ones (e.g., reward prediction error calculation) [28].

Different NF protocols treat states rather differently. An explicit-intermittent protocol (Figure 3A) temporally separates the high-level state for regulation (x_{regulate}), of performing conscious intentional actions (termed mental strategies) from the state of reward presentation and evaluation of action utilities (x_{feedback}). This poses a temporal credit assignment problem (Figure 2B). Since participants get limited information about the relationship between mental/neural states and actions during regulation, they might achieve the desired brain pattern wastefully: a neural analog of superstitious behavior in animal conditioning paradigms [29]. However, dissociating ‘regulation’ and ‘feedback’ states could have benefits such as simplifying concentrating on a mental strategy by removing the need to track feedback continually.

Conversely, an explicit-continuous NF protocol demands that participants perform a conscious action (i.e., mental strategy) and evaluate the report of its utility simultaneously (Figure 3B, $x_{\text{regulate+feedback}}$). Although the immediate report of utility greatly simplifies the temporal credit assignment problem (albeit not solving it completely, as measurement-related delays still exist), it also divides the participant’s attention between the mental strategy performance and continuously tracking the feedback, and can exacerbate the misbehavior problem mentioned earlier.

This analysis of states sheds important light on the choice of actions during explicit NF practice. In common NF practice the protocol does not dedicate a state for participants to choose which mental strategy to employ. Given that a choice ‘must occur’ in every NF cycle, participants might make it immediately following feedback presentation, during the ‘rest’ state, or even after

the onset of the regulation phase. This may induce variability; it could mean that choice, which is the central operation in RL, is performed with haste, and it could even become conflated with other protocol states. Thus, the absence of a distinct task state for participants' choice may turn out to be an unfortunate practice that confounds explicit NF protocols. It could therefore be desirable to add a 'choice' block at the beginning of every explicit NF cycle (Figure 3A,B, x_{choice}), providing a distinct time period for participants to select the mental strategy they wish to employ in the subsequent regulation phase. In implicit protocols (Figure 3C), where participants are not instructed to apply mental strategies, such conflation between states is avoided at the cost (or benefit) of discarding this aspect of NF training (see the following section).

Crucially, learning about the utility of different lower-level internal states (and actions) in explicit and implicit protocols demands their mental representation in some form by the agent [30]. Hence a central part of NF learning would be the representation of internal neural/mental states that the participants transition between during training; this also involves metacognition (Box 1).

Box 1. Representation learning and metacognition in NF training

Metacognition and representational learning play several roles in NF, in determining internal characterizations of cognitive and neural processes, and in governing regulation learning. We consider four roles: awareness of the relevant components of internal activity, representational redescription of the internal activity, confidence in the choice or in the prospect of reward, and internal models of the transition function $f[x_n(t), A_n(t)]$ that determine the effect of actions and the reward process.

Becoming aware of at least some facets of the changing neural activity one is trying to regulate is considered in the NF literature as an additional learning process, namely awareness learning [7,56]. Some training protocols have attempted to teach participants to develop this awareness, with or without recommending active manipulation [57–60]. In our RL framework, developing such an ability would be a form of explicit representation learning of internal lower-level states that report on changes in target neural patterns. Implicit protocols may limit such explicit representation learning during NF practice, as trainees are not instructed to search for contingencies between their mental states and the sensory feedback. Additionally, temporally rich sensory feedback in explicit continuous protocols (of rewarding desired changes or of their anticipatory neural states) may particularly facilitate such representational learning (see Outstanding questions), and also the learning of internal models (see later).

Representations (or perhaps augmented representations) of the state can have a substantial impact on the speed of learning. This is because the internal neural activity is very high-dimensional, and so can beneficially be filtered [61] to determine components that are associated with transitions and rewards. More generally, good representations afford appropriate generalization, so values and policies can be useful even if some aspects of the neural activity on a trial differ from that on previous trials, for example, because of noise.

The most classic form of metacognition concerns the confidence that an individual has about the relationship between mental states and actions and the neural target patterns. Confidence plays a role in action selection [62], strategies for exploration [63,64], and in regulating the course of learning. In a study employing an online brain–computer interface (BCI) procedure during an RL gambling task, several confidence measures were associated with individuals' action selection as well as other learning parameters [18]. Another study probed the ability of subjects to predict neural patterns in the regulated region and their confidence regarding those predictions (in periods when feedback was not presented), demonstrating correlations between prediction accuracy and confidence [60]. Similarly, in NF contexts, confidence regarding the utility of internal actions (e.g., mental strategies) and states may influence learning dynamics.

Finally, individuals may build an internal model of their environment (such as a cognitive map of a maze [27,65]), which characterizes the transitions $f[x(t), A(t)]$ between states dependent on actions and the potential consequent reward $r(t)$. This model can be used for planning to increase reward. In the case of NF, even a coarse model might be useful for conscious control: for example, of the effect of keeping 'calm' or of imagining the performance of some particular class of external actions in order to control internal activity.

Actions: causal precursors of the reinforced neural patterns

The second critical component of an RL problem is the actions. In much of the traditional behavioral literature, actions were typically observable movements (e.g., lever presses, stimulus choices, spatial navigation, motor sequence performances, etc.). More recently, however, internally directed actions – such as the deployment of working memory, attention, or cognitive control – have been increasingly considered in RL contexts [31–33]. Consistent with such an extension, in NF the actions taken by participants in order to accumulate rewards may be intentional, coincidental, or habitual, but are intended to be purely mental and/or neural (for movement confounds see [34,35]), performed via thought, imagery, or other mental/neural processes.

As for states, the structure of actions in NF differs according to protocol type. NF actions may be conscious or unconscious, depending partly on the explicitness of the instructed regulation goals (Figure 3A–C, Actions). In explicit protocols, actions are primarily the mental strategies that participants explore [usually fixed for a trial through task instructions, and thus denoted by the constant $A_n(t) = A_n$]. The space of actions participants explore (Figure 2C) in order to influence the feedback in a desired manner may be limited only by their individual capacities (i.e., their cognitive resources, imaginative capabilities, etc.), or it can be restricted or expanded through given task instructions (e.g., to ‘retrieve positive memories’, ‘relax’).

While explicit protocols drive participants towards such intentional mental actions, other implicit mental/neural actions may participate in learning. In fact, the high-level category of a mental strategy that is deployed in seconds-long time frames or more is constructed by more intricate mental/neural processes denoted by:

$$A_n(t) = A_n + a_n(t) + \varepsilon_n^a(t) \quad [2]$$

where $a_n(t)$ is a possibly subconscious strategy/action, a mental or neural control process that determines the transitions between low-level internal states (Equation 1), and $\varepsilon_n^a(t)$ is action noise. Then, even a good overall intentional strategy, A_n , can be systematically adjusted via $a_n(t)$; this may explain learning effects that exceed the intentional use of strategies. In implicit protocols (Figure 3C), unless participants develop an explicit regulation goal by themselves, actions could not be intentionally chosen mental strategies that are applied to influence the feedback, but only such fine-grained mental/neural control processes. In either case, actions could be defined as any mental, neural, or other physiological causal precursor of the reinforced neural activity patterns that determine the feedback.

It has been proposed that conscious efforts to apply mental strategies, which are minimized in implicit protocols, could have advantageous or detrimental effects on regulation learning, depending on the context [12,36]. Explicit NF protocols are assumed to provide participants with effective hints about good strategies; however, inaccurate hints may impede neural regulation. Thus, this assumption should be tested empirically, that is, assessing whether strategy application in general, and certain mental contents in particular, are in correspondence with the gradual change in neural patterns in the reinforced direction (see Outstanding questions). However, in explicit NF studies, mental strategies are rarely documented systematically (for every NF cycle of the practice) or taken into consideration in analyses (for notable exceptions see [14,37–39]). As noted, the lack of a protocol-mandated ‘choice’ epoch in a trial in typical NF protocols makes such documentation difficult. Crucially, introducing a choice epoch may be exploited for documentation of mental strategies as well. In Box 2 we discuss the significance and possible modes of mental strategy documentation and characterization.

Box 2. Documentation and characterization of mental strategies

During explicit NF regulation phases, participants employ mental strategies in order to influence the feedback. A strategy involves one or several perceptual, affective, imaginative, cognitive, or metacognitive components. Explicit protocols differ in instructions for strategy application. While some studies do not mention preferable contents [5,66], most direct participants towards process-specific thoughts/imagery (e.g., emotion regulation) [11,14,67]. The cues depicted on the feedback interface may also tilt participants towards specific strategies [5,68,69]. Nonetheless, the space of options of mental strategy application remains considerable, leaving participants substantial choice (see Figure 2C in main text).

Mental strategies are not often documented and assessed by investigators. When they are, this is usually performed at the end of a practice session, following a few dozen regulation trials. Thus, confounds associated with memory and cognitive abilities, regulation success, or other interface-specific effects may arise, implying that participants' choices of mental strategies should rather be documented in real time. To keep practice interferences minimal, we suggest combining predefinition of strategies before each NF run (several successive trials), logging of strategy selection during choice screens, and phenomenological characterization of strategies at the end of the session, via designated questionnaires.

Logging could be arranged during a choice condition in which participants select via a keypress their chosen strategy out of several predefined possibilities (see Figure 3A,B in main text, first task epoch). Importantly, predefining possible strategies before each NF run would not restrict participants to a defined set of strategies along the session, as options may be updated between runs, and could always include an 'other' set option that enables a free choice, which may be characterized following practice, thereby minimizing memory-related confounds. Lastly, introducing a choice screen along with predefining strategies before each run may considerably improve learning by allowing participants to conceive their strategies once every several trials in a temporally unrestricted manner.

After logging strategy selection, we suggest performing a phenomenological characterization. Previous works have characterized mental strategies in various qualitative ways [70–72], including eliciting a free verbal description of each strategy. Nonetheless, quantitative measuring tools are lacking. In other fields, attempts to characterize mental states quantitatively have been pursued [73]. Similarly, in NF, each strategy could be rated by the participants themselves according to known psychological dimensions, such as arousal and valence, as well as additional phenomenological categories. For example, a strategy of retrieving of a specific personal positive memory may be classified as imaginative rather than sensory, as involving visual or tactile imagery contents, as relating to self as opposed to others, and so forth. Importantly, such a characterization may be confounded by the ability of subjects to identify internal states and to verbally report them (e.g., in case of alexithymia, which relates to deficient cognitive processing of emotions); to account for this potential confound, such traits could be assessed via designated questionnaires [74]. Further work should develop such exhaustive classification specifically designed to characterize mental strategies.

Rewards: evaluation of the sensory feedback

The third central element in RL is the reward. In conventional RL there are two critical outcome quantities: the immediate reward $r_n(t)$ at time t on trial n , associated with state $x_n(t)$ and action $A_n(t)$, and then the long-term return which defines the success of a trial. Typically, the agent considers the return in a prospective manner, as a simple expected value (denoted by E) of a sum of future rewards from time t across the remainder of a trial:

$$R_n(t) = E[\sum_{u \geq t} r_n(u)] \quad [3]$$

where u indicates future time steps in the current regulation epoch, although some NF protocols include various sorts of nonlinearity (e.g., feedback thresholding [40,41]).

In our notation, at least ignoring the time taken for data processing, the immediate reward can often be written as function of the state:

$$r_n(t) = g[x_n(t)] + \varepsilon_n^r(t) \quad [4]$$

where g , the *measurement function*, is determined by the experimenter's choice of where (from which brain region) and how (e.g., with which recording method, and according to which metric) to monitor neural activity, and $\varepsilon_n^r(t)$ is reward noise.

In terms of reward timing: in continuous protocols the information regarding $r_n(t)$ is provided in sensory form. In intermittent protocols, as long as $t \neq T$ (T indicating the end of the regulation epoch), $r_n(t) = 0$.

As regulation terminates, the participant receives a single apparently immediate reward $r_n(T)$ that accounts for the past regulation epoch, leaving participants to solve the temporal credit assignment problem, as discussed earlier (Figure 2B), and therefore may impose extra cognitive load. Importantly, certain recording methods may limit the fidelity of the temporal contingency between rewarding feedback and neural/mental states and actions. While electroencephalogram (EEG)-NF enjoys high temporal resolution, longer delays in fMRI-NF practice imply that the temporal credit assignment problem for continuous protocols remains far from being eliminated.

In most cases [e.g., blood oxygenation level-dependent (BOLD) fMRI-NF] the reward is measured by the match between desired and actual target activity during a ‘regulation’ epoch relative to that during a ‘rest’ epoch, making

$$r_n(t) = g[x_n(t)] + \varepsilon_n^r(t) - g(x_{rest}) \quad [5]$$

in which case participants could optimize the reward by minimizing $g(x_{rest})$ rather than maximizing $g[x_n(t)]$; hence rest periods are also potentially valenced, either implicitly or possibly through some conscious action specific for rest periods, A_{rest} (Figure 3A,B, Actions). Moreover, continuously evaluated sensory feedback, which does not induce substantial credit assignment problems, is the most straightforward match for standard RL, as opposed to intermittent feedback. By contrast, intermittent feedback has the technical advantage of averaging over what can be substantial noise, therefore providing more accurate and trustworthy feedback.

Regarding the explicitness of regulation: in explicit protocols, the sensory feedback provided by the environment (i.e., the feedback interface) is evaluated by participants in the face of an instructed regulation goal (e.g., try to make the sensory feedback increase). As a result, the sensory feedback functions as a reward when in line with the instructed regulation goal, and as punishment when it deviates from it. However, as noted earlier (Box 1), sensory feedback could potentially depict additional characteristics of the neural pattern that are not directly associated with reward, but nevertheless help participants understand how different conscious mental states and actions alter their neural patterns as reflected by the feedback interface. Importantly, such knowledge may promote regulation learning [42] (see Outstanding questions). In implicit NF, if participants do not deduce a regulation goal, the sensory feedback is entirely evaluative, that is, it does not provide potentially useful information regarding the link between the participants’ mental states and the neural dynamics. This may lead to a hard statistical problem of relating a simple, scalar, reward signal to the booming, buzzing, confusion of the extremely high-dimensional neural dynamics.

RL algorithms for NF

RL theory deals with classical as well as instrumental conditioning. Previous theoretical works have suggested that NF learning could be based on classical conditioning mechanisms [40,43,44]. However, in classical conditioning the execution of an action (i.e., a conditioned response) is incidentally tied to the prediction of reward or punishment. Hence, in the context of self-regulation learning via NF training, this will translate only to instances in which feedback presentation by itself elicits the desired action (e.g., upregulation of a neural pattern), that is, when it is measured from prediction-related brain regions (e.g., the ventral striatum).

Instrumental conditioning, which could explain all other NF training instances, enjoys two main classes of action selection algorithms, model-based and model-free, corresponding to deliberative goal-directed behavior and habitual behavior, respectively [45,46]. In model-based RL (MB-RL), the value of actions is computed by mentally simulating the consequences of possible

Box 3. Algorithmic descriptions of NF learning

Consider applying strategy/action $A_n(t) = A_n = A$ constantly during the ‘regulate’ epoch in cycle n of an explicit-intermittent protocol. The action’s average reward $Q(x_{\text{regulate}}, A)$ can be acquired via Q-learning (C. Watkins, PhD Thesis Cambridge University, 1989) [13,75], with update:

$$Q(x_{\text{regulate}}, A_n) \leftarrow Q(x_{\text{regulate}}, A_n) + \alpha \delta_n \quad [\text{I}]$$

where α is the learning rate, and δ_n is the prediction error:

$$\delta_n = r_n(T) - Q(x_{\text{regulate}}, A_n) \quad [\text{II}]$$

with $r_n(T)$ as the reward on the terminated regulation epoch.

Then, a policy $\pi(x_{\text{regulate}})$ converts $Q(x_{\text{regulate}}, A)$ into the choice of A (e.g., via softmax):

$$P(A_n = A; x_{\text{regulate}}) = \pi_A(x_{\text{regulate}}) = \frac{\exp[\beta * Q(x_{\text{regulate}}, A)]}{\sum_B \exp[\beta * Q(x_{\text{regulate}}, B)]} \quad [\text{III}]$$

where β is the inverse temperature determining how exclusively $\pi(x_{\text{regulate}})$ focuses on exploiting $\arg\max_A Q(x_{\text{regulate}}, A)$, and B sums over all possible strategies. Thus β regulates the extent of exploration of apparently suboptimal options. Such a scenario is known as a multiarmed bandit problem [76].

Together, algorithms I–III could capture neural regulation learning through the intentional use of mental strategies, in explicit-intermittent and, to some extent, explicit-continuous protocols, with high-level actions and states as intentional mental strategies and representations of task epochs, respectively. However, to account for the fine-grained internal actions and states unfolding in explicit and implicit protocols, we must consider the broader range of RL algorithms described in detail by Sutton and Barto [13]. Consider, for example, the simple actor–critic learning rule [47]. This starts from the value $V[x(t)]$ of state $x(t)$, namely the expected return E across the regulation epoch starting from state $x(t)$:

$$V[x(t)] = E\left[\sum_{u=t}^T r(u)\right] \quad [\text{IV}]$$

summing the reward expected from time t until the end of the epoch T (dropping the previous cycle index n). This averages over the detailed choices of subsequent actions $A(u) = A + a(u) + \varepsilon^a(u)$ determined by the policy $\pi[x(t)]$. In fully intermittent protocols, $r(t) = 0$ while $t \neq T$.

Learning $V(x(t))$ depends on its parameterization, and how those parameters change. Typically, $V(x(t))$ is written as:

$$V(x(t)) = \sum_i \phi_i(x(t)) \omega_i^V \quad [\text{V}]$$

where $\phi_i(\cdot)$ are internal representations of the neural activity and ω_i^V are learned weights [one for each $\phi_i(\cdot)$]. As noted in Box 1, good representations influence the ease of learning $V(x(t))$. The weights are changed according to the temporal difference learning rule:

$$\omega_i^V = \omega_i^V + \alpha \delta(t) \phi_i(x(t)) \quad [\text{VI}]$$

where α is the learning rate and $\delta(t)$ is the temporal difference [77] prediction error at time t :

$$\delta(t) = r(t) + V(x(t+1)) - V(x(t)) \quad [\text{VII}]$$

In the actor–critic rule, the policy’s choice of $A(t)$ is also criticized by $\delta(t)$; the more positive it is, the better $A(t)$, based on an immediate reward $[r(t) \gg 0]$ or because the state changed in a direction that favors future reward $\{V[x(t+1)] \gg V[x(t)]\}$. The policy can be parameterized using its own weights ω_j^{π} in a similar way to $V(x(t))$, with the choice of A , $a(t)$, and $\varepsilon^a(t)$ being determined by the equivalent of the softmax for the constant, explicit-intermittent case. Through the learning processes described by algorithms IV–VII, specific neural activity patterns in certain brain regions (states) and the determinative control processes that bring them about (actions) may become represented and weighted (i.e., valenced) according to the ongoing evaluative feedback ($r(t)$). Such algorithms could capture regulation learning either along with the use of mental strategies in explicit protocols, or with no instructions for strategy application whatsoever, that is, in implicit protocols.

Future studies could utilize these algorithmic suggestions in order to quantify the two potential kinds of learning in explicit and implicit protocols, and thus assess to what extent does intentional use of mental strategies matter for regulation, and which unconscious internal states and actions determine regulation success (see Outstanding questions).

sequences of actions that transition the agent through different states within an internal model of the world. This model includes the world states, the action-dependent transition probabilities between them, and the expected rewards for each action in each state [45].

There may be some room for MB control in NF: for example, in a coarse understanding of the consequences of particular modes of thought (e.g., expectations about ‘clearing one’s mind’) (Box 1). However, intentional NF actions (i.e., mental strategies) are extremely diverse, and even in the explicit-continuous case, NF experiments are far too short to afford a substantial opportunity for participants to learn exhaustively about the complex relationship between their choices and detailed aspects of their neural target activity. Similarly, the structure of implicit NF tends to discourage the learning process from being goal-directed. Nevertheless, one cannot rule out the possibility that participants might construct an MB problem/solution for themselves.

It nevertheless remains more likely that many aspects of both explicit and implicit NF learning and performance can be better described through model-free RL (MF-RL) algorithms. In MF-RL, the values of actions, that is, the ‘return’ expected from executing those actions, or possibly just the policy, are learned directly from experience, via trial-and-error interaction with the environment. Specifically, agents can learn about the expected utility of different states and use this to criticize an explicit policy (as in actor–critic algorithms [47]), or they can learn the utility of each specific action in each state (as in Q-learning [13], C. Watkins, PhD Thesis, Cambridge University). In Box 3, we describe how each of these MF algorithmic solutions may be aligned with two different aspects of NF training.

Concluding remarks and future perspectives

In RL terms, the major puzzle underlying NF is its efficiency. Animal NF studies have shown that it is possible to boost the activity of a single neuron whilst suppressing the activity of neighboring neurons, for essentially arbitrary choices of neurons, in a reasonable training period [48,49]. This seems an extraordinarily hard search and exploration problem, and how it is solved remains unknown. Human NF paradigms currently operate with far less precision, with still some substantial evidence of neural regulation learning effects [5,8,9,79].

In the present opinion article we have attempted to set the stage for a solution to this puzzle by examining the practice of NF under the lens of RL. Our characterization licenses comparisons between typical NF protocols in terms of their unique advantages and limitations, and puts in sharper perspective unattended aspects of NF learning – such as credit assignment (Figure 2B) and exploration of actions (Figure 2C) – that are extensively investigated in RL.

Several previous studies have considered certain aspects of NF as an RL problem [19–21,50], or have alternatively applied control theory to shed light on NF training [42,51–53]. The framework we outlined adopts similar goals in providing a firmer computational and algorithmic footing for NF training. However, we would argue that there are two distinct characteristics of the current perspective. The first is the general and detailed scope of modeling that accounts for the variety of NF protocol types (including the less common explicit-intermittent and pure implicit protocols). For each protocol type, we have highlighted its unique advantages and constraints in the context of RL. A second key aspect of the proposed framework is the formulation of two testable algorithmic suggestions (Box 3) that describe two complementary learning processes in NF training, namely, the development of behavioral policies for application of mental strategies during explicit protocols, and regulation based on fine-grained neural states and actions, that applies to explicit as well as implicit protocols.

Based on this proposal, several practical recommendations can be made for better planning and execution of NF trials. First, the definition of NF actions as determinative precursors of the reinforced neural pattern may guide future investigations towards causal rather than correlative inspection of the link between actions and rewards (see Outstanding questions). The influence of intentional actions (i.e., mental strategies) on regulation success could be assessed quantitatively (Boxes 2 and 3), or strategy application could be manipulated experimentally via differing task instructions. To

Outstanding questions

What is the relationship between actions and rewards in NF practice? In explicit NF protocols, do specific modes of thought correspond with regulation success in general? Which mental contents contribute uniquely to regulation of specific brain areas? In implicit NF protocols, which neural actions play a causal role in regulation success?

Which regions in the brain play causal roles in particular mental processes? Could manipulating these brain regions via NF bring forth new insights about these mental processes?

Which metacognitive and representational processes take part in NF training? How do these processes interact with regulation learning?

Could richer feedback interfaces improve regulation abilities by depicting not only neural patterns in a target brain region but also anticipatory neural states? Could such anticipatory states be identified based on theory-based assumptions, or by data-driven approaches?

How could the various RL parameters be assessed over the course of practice? Can these parameters then be exploited for an optimized, personal, NF experimental design?

Which kinds of measurement functions $g[x_i(t)]$ are optimally suited for endowing participants with the ability to regulate a neural mechanism underlying a specific aberrant mental process? Which other RL parameters of NF practice relate to symptoms alleviation?

examine causal links between unconscious neural actions and appropriate regulation, transcranial magnetic stimulation could be used. Alternatively, implicit NF protocols that restrict the use of mental strategies could be exploited for delineating such causal links [36]. Second, as elaborated in Box 1, metacognition plays a number of potential roles in NF learning, such as the development of awareness to the changing neural activity, or the confidence regarding the utility of internal actions and states. Measurement of such metacognitive elements is sorely needed: for example, in the case of explicit-intermittent protocols, by asking participants to predict their neural activity and rate their confidence regarding such predictions following regulation and prior to feedback presentation [60], or by assessing metacognitive skills with designated tasks and questionnaires [54,55]. Third, our analysis recognizes a systematic confound in explicit NF protocols, namely, the lack of a distinct task state dedicated for the choice of intentional actions, as well as the neglect of the characterization of these actions. We suggest the use of a ‘choice’ screen to log participants’ choices and to characterize them in quantifiable ways. Further work is needed to develop such tools.

Finally, we did not elaborate on an important element of NF practice, namely, the experimenter’s choice of a measurement function $g[x_n(t)]$ of neural target activity (see Outstanding questions). Indeed, NF training success and clinical utility depends considerably on psychophysiological accurate measurement functions. We believe that the formal description devised here of their role, along with other critical elements, paints a clearer picture of the process of regulation learning via NF. Thus, we hope it will promote NF both as a basic science method to investigate the link between mental processes and neural activity and as an efficient neuropsychiatric intervention.

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Declaration of interests

T.H. is a chief medical scientist of Graymatters Health. The other authors declare no conflicts of interest in relation to this work.

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